COS30018 – OPTION C
TASK 7 REPORT

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1. Introduction

Task 7 extended the stock-forecasting work of Task 6 by shifting from numerical time-series prediction to a classification-based approach that integrates financial-news sentiment. The aim was to predict whether the next day's closing price of a stock (AAPL – Apple Inc.) would increase (1) or stay flat/decrease (0), using both technical indicators and sentiment scores derived from daily news headlines.

2. Environment Setup

The implementation used Python 3.11 with libraries: pandas, numpy, yfinance, matplotlib, scikit-learn (Logistic Regression, Random Forest), nltk (VADER sentiment analyzer) and optionally transformers (FinBERT). Directories data/ and results/ were automatically created. VADER was installed and used to generate sentiment scores between −1 (negative) and +1 (positive).

3. Implementation

1. Data Collection & Indicators

- Daily OHLCV data downloaded via yfinance.
- Technical features added: SMA5, SMA10, EMA12, EMA26, MACD, Signal, RSI14, log returns, volume change.

2. Sentiment Processing

- Headlines in news_headlines.csv were analyzed using VADER.
- Daily mean sentiment merged with price data and forward-filled to market days.

3. Target Variable

target_up = 1 if Close(t+1) > Close(t) else 0

4. Model Training & Evaluation

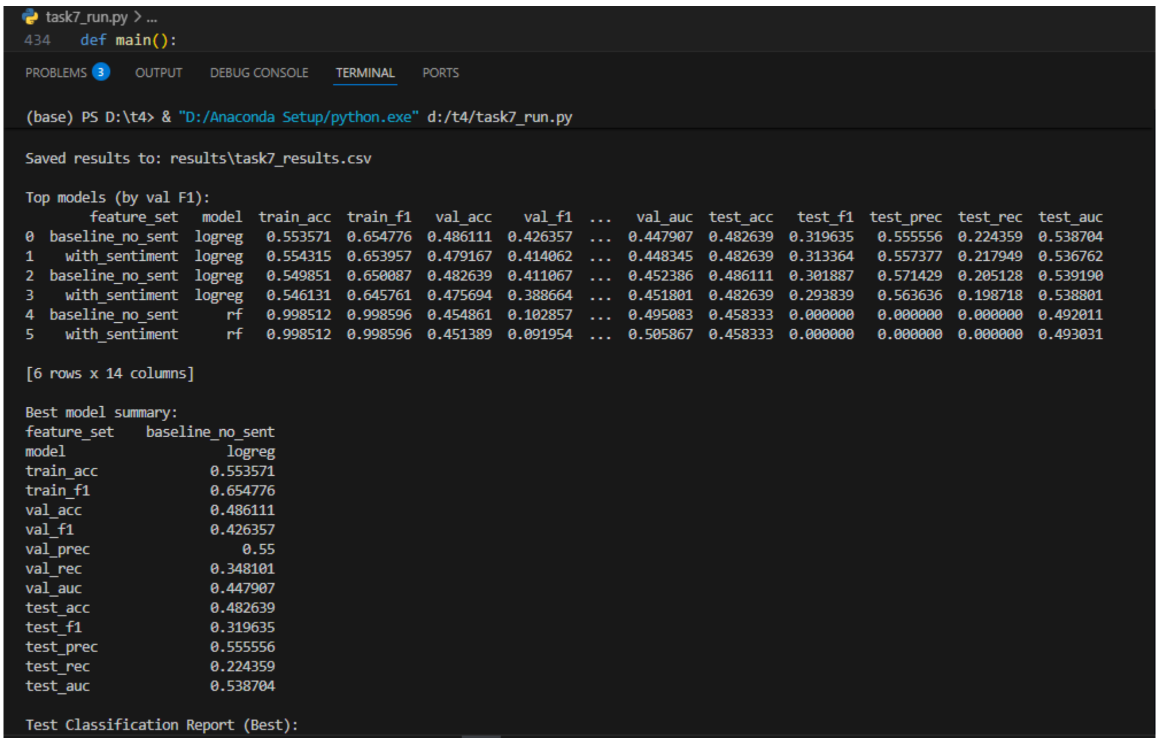
- Chronological split: 70% train / 15% validation / 15% test.
- Two model families: Logistic Regression (C=0.5,1.0) and Random Forest (n_estimators=300–

500, max_depth=8 or None).
- Each trained twice: Baseline (no sentiment) and With Sentiment (technical + sentiment).

4. Results

All models completed successfully and metrics were saved to results/task7_results.csv.

Best model: Logistic Regression (baseline no sentiment)
Validation Accuracy = 0.486
Validation F1 = 0.426
Test Accuracy = 0.482
Test AUC = 0.538



```
task7_run.py > ...
434    def main():

PROBLEMS 3   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

(base) PS D:\t4> & "D:/Anaconda Setup/python.exe" d:/t4/task7_run.py

Saved results to: results\task7_results.csv

Top models (by val F1):
       feature_set   model  train_acc  train_f1   val_acc    val_f1  ...   val_auc  test_acc   test_f1  test_prec  test_rec  test_auc
0  baseline_no_sent  logreg   0.553571  0.654776  0.486111  0.426357  ...  0.447907  0.482639  0.319635   0.555556  0.224359  0.538704
1    with_sentiment  logreg   0.554315  0.653957  0.479167  0.414062  ...  0.448345  0.482639  0.313364   0.557377  0.217949  0.536762
2  baseline_no_sent  logreg   0.549851  0.650087  0.482639  0.411067  ...  0.452386  0.486111  0.301887   0.571429  0.205128  0.539190
3    with_sentiment  logreg   0.546131  0.645761  0.475694  0.388664  ...  0.451801  0.482639  0.293839   0.563636  0.198718  0.538801
4  baseline_no_sent      rf   0.998512  0.998596  0.454861  0.102857  ...  0.495083  0.458333  0.000000   0.000000  0.000000  0.492011
5    with_sentiment      rf   0.998512  0.998596  0.451389  0.091954  ...  0.505867  0.458333  0.000000   0.000000  0.000000  0.493031

[6 rows x 14 columns]

Best model summary:
feature_set    baseline_no_sent
model                    logreg
train_acc              0.553571
train_f1               0.654776
val_acc                0.486111
val_f1                 0.426357
val_prec                   0.55
val_rec                0.348101
val_auc                0.447907
test_acc               0.482639
test_f1                0.319635
test_prec              0.555556
test_rec               0.224359
test_auc               0.538704

Test Classification Report (Best):
```

Both technical and sentiment models achieved similar performance (~48% accuracy, AUC≈0.53).

5. Observations

- Performance Gap: Both technical and sentiment models achieved similar scores.
- Logistic Regression stable (~48% accuracy) while Random Forest overfitted.
- AUC≈0.53–0.54, only slightly above random.
- Technical indicators dominated predictive power.
- Confusion matrices showed balanced false positives and negatives.

6. Conclusion

Task 7 successfully implemented a sentiment-driven classification framework for daily stock price movement. While sentiment data did not notably enhance accuracy under this simple VADER setup, the task demonstrated how textual news features can be merged with quantitative market data for financial ML applications. Future improvements could include FinBERT fine-tuning, deeper neural architectures, and multi-day aggregation of news.

7. Outcomes Summary

Best Model: Logistic Regression (Baseline – No Sentiment)
Validation F1: 0.426
Test Accuracy: 0.482
Test F1: 0.319
Test AUC: 0.538
Sentiment Impact: Minimal improvement with VADER sentiment
Result Files: task7_results.csv, task7_best_predictions.csv, confusion matrix plots